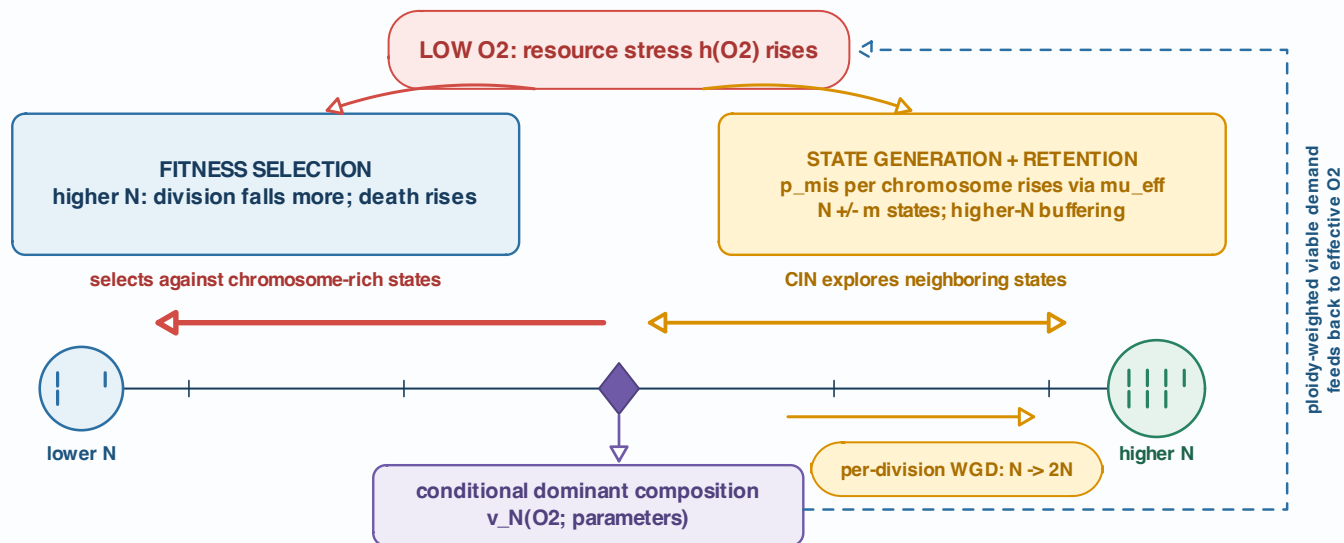


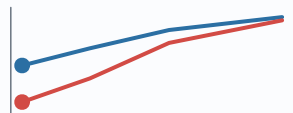
## A Resource limitation couples opposing pressures on ploidy evolution



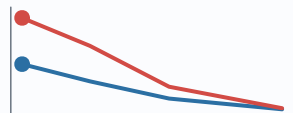
## B Low- $O_2$ fitness selection

— lower N — higher N

Effective division rate



Hypoxia-associated death

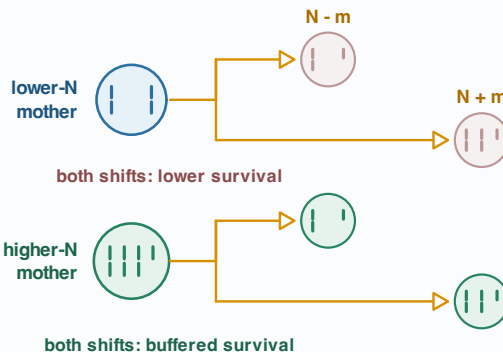


O2 increasing

Low  $O_2$  favors lower-N expansion

## C State generation + survival filtering

$p_{\text{mis}}$  per chromosome rises  
via  $\mu_{\text{eff}}$



WGD:  $N \rightarrow 2N$   
constant per division

out-of-grid offspring  
enter nonviable pool

## D Population integration

fitness + CIN/WGD + survival

fixed- $O_2$  operator  
 $M(O_2)$

conditional dominant composition  $v_N$



chromosome state N

mean ploidy =  $\sum(N v_N) / 22$

interpret when the spectral gap resolves  $v_N$